Continuous Fourier Transform Explorer

An interactive tool for Fourier-transform pairs, the Duality Property and the Area Property — project report

Item	Details
Course	Signals and Systems
Project code	A43 — Continuous Fourier Transform Explorer (GUI)
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Deliverable	Interactive web application (single HTML file) and this report

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Abstract

This report documents the theory, design and verification of the **Continuous Fourier Transform Explorer**, an interactive web application built for the Signals and Systems course (project A43). The user selects one of five standard signals (rectangular pulse, triangular pulse, Gaussian pulse, sinc and one-sided decaying exponential), adjusts its amplitude, width and time shift, and sees the magnitude, phase, real part and imaginary part of its Fourier transform update in real time.

Two analytical properties are built in as live demonstrators. The **Duality Property** shows that if $x(t) \leftrightarrow X(f)$ then $X(t) \leftrightarrow x(-f)$; both transform pairs are drawn side by side and the second pair is confirmed by a numerical Fourier transform. The **Area Property**, $X(0) = \int x(t)dt$ and $x(0) = \int X(f)df$ (equivalently $(1/2\pi)\int X(\omega)d\omega$), is evaluated numerically and compared with the closed-form values. In a sweep of 240 parameter combinations the area errors stayed below 1 % of the reference scale and the duality gap stayed below 6.3 % of the peak value, the larger figure arising only for very narrow pulses where the finite integration window blurs the edges.

1 Introduction

1.1 Background

The Fourier transform expresses a signal as a continuous superposition of complex exponentials. It converts a description in time into a description in frequency and is the basis of filtering, modulation, sampling theory and spectral analysis. For students the difficulty is usually not the integral itself but building intuition: how a narrow pulse becomes a wide spectrum, why a time shift changes only the phase, and why transform pairs come in mirrored couples. An interactive tool lets those relationships be seen rather than only calculated.

1.2 Problem statement

The project brief asks for a graphical interface in which the user selects a signal (rect, Gaussian, triangle, sinc, exponential), adjusts its parameters, and sees $|X(f)|$, $\angle X(f)$, Re X(f) and Im X(f) update in real time. Two extras are required: a demonstrator for the Duality Property with both transform pairs shown side by side, and an implementation and verification of the Area Property.

1.3 Objectives

- Provide the five required signals with adjustable amplitude and width, plus a time-shift control that makes the phase and the real and imaginary parts informative.
- Display magnitude, phase, real and imaginary parts of the transform and refresh them immediately when any control changes.
- Demonstrate duality by drawing $x(t) \leftrightarrow X(f)$ and $X(t) \leftrightarrow x(-f)$ together and verifying the second pair numerically.
- Verify both halves of the Area Property numerically and report the agreement.
- Support both the Hz form and the rad/s form of the transform so the $1/2\pi$ and 2π factors are visible and correct.
- Include viva material, in particular the transform of a constant derived from $\delta(t) \leftrightarrow 1$ through duality.

1.4 Deliverables

The deliverables are a single self-contained HTML page (no installation, runs in any modern browser), this report, and a companion guide that explains the page component by component and walks through the source code.

2 Theoretical background

2.1 Definition and conventions

Two conventions are in common use. The first uses ordinary frequency f in hertz and is symmetric, with no 1/2π factor. The second uses angular frequency ω in radians per second and puts a factor 1/2π in the inverse transform. The application supports both through a toggle; the two are linked by X(ω) = X(f) evaluated at f = ω/2π.

Table 1. The two conventions side by side. All integrals run from −∞ to ∞.

Item	Hz form (f)	Rad/s form (ω)
Analysis (forward)	X(f) = ∫ x(t) e^(−j2πft) dt	X(ω) = ∫ x(t) e^(−jωt) dt
Synthesis (inverse)	x(t) = ∫ X(f) e^(j2πft) df	x(t) = (1/2π) ∫ X(ω) e^(jωt) dω
Duality	X(t) ↔ x(−f)	X(t) ↔ 2π·x(−ω)
Area property	X(0) = ∫x dt, x(0) = ∫X df	X(0) = ∫x dt, x(0) = (1/2π)∫X dω
Constant and impulse	δ(t) ↔ 1, 1 ↔ δ(f)	δ(t) ↔ 1, 1 ↔ 2π δ(ω)

> **Note on the brief.** The area formula x(0) = (1/2π)∫X(f)df in the project description is the rad/s form. In the Hz form the factor 1/2π is absent. The application therefore shows whichever form matches the selected axis.

2.2 Existence

A sufficient condition for the transform to exist is that x(t) is absolutely integrable and has a finite number of discontinuities and extrema in any finite interval (the Dirichlet conditions). All five signals in the application satisfy this, including the sinc, whose transform is a rectangle. Signals such as the constant 1 violate absolute integrability, so their transforms exist only in the generalised sense, through the impulse δ.

2.3 Properties used in the project

- **Linearity:** $a \cdot x_1(t) + b \cdot x_2(t) \leftrightarrow a \cdot X_1(f) + b \cdot X_2(f)$. The amplitude control A scales X(f) by the same factor.
- **Time shift:** $x(t - t_0) \leftrightarrow X(f)\, e^{(-j2\pi f t_0)}$. Magnitude is unchanged, phase gains the linear term $-2\pi f t_0$, and the real and imaginary parts are rotated into each other.
- **Time scaling:** $x(at) \leftrightarrow (1/|a|)\, X(f/a)$. Narrowing a pulse widens its spectrum. This is the time-bandwidth trade-off seen when σ, T or B are changed.
- **Duality:** described below.
- **Area:** described below.

2.4 The Duality Property

Statement. If x(t) ↔ X(f), then X(t) ↔ x(−f). In rad/s form, X(t) ↔ 2π·x(−ω).

Proof. Start from the synthesis equation x(t) = ∫ X(f) e^(j2πft) df. Interchange the names of the two variables: x(f) = ∫ X(t) e^(j2πtf) dt. Replace f by −f: x(−f) = ∫ X(t) e^(−j2πft) dt. The right-hand side is exactly

the Fourier transform of the function X(t), so X(t) ↔ x(−f). In the rad/s form the synthesis integral carries 1/2π, which becomes the factor 2π in the result.

Example. rect(t) ↔ sinc(f). Duality says sinc(t) ↔ rect(−f) = rect(f), because the rectangle is even. The sinc signal in the application is therefore obtained without any new integration: a sinc of bandwidth B transforms to (1/B) rect(f/B).

2.5 The Area Property

Setting f = 0 in the analysis integral gives **X(0) = ∫ x(t) dt**: the value of the spectrum at zero frequency is the net area under the signal. Setting t = 0 in the synthesis integral gives **x(0) = ∫ X(f) df** (or (1/2π)∫X(ω)dω): the value of the signal at t = 0 is the net area under the spectrum. The two statements are dual to each other.

> **Jump discontinuities.** If x(t) jumps at t = 0, the inverse transform converges to the midpoint of the jump. For the exponential A·e^(−at)·u(t) with t_0 = 0 the application therefore finds x(0) = A/2, not A. The same happens for a rectangle whose edge is placed at t = 0.

3 Signal catalogue and closed-form transforms

Each signal is implemented with a closed-form transform, so the plots are exact (no numerical error) and the numerical routines are used only for verification. The sinc function is defined as sinc(u) = sin(πu)/(πu) with sinc(0) = 1. Every signal can be delayed by t_0, which multiplies its transform by e^(−j2πft_0).

Table 2. The five signals and their transforms.

Signal x(t)	Parameters	Transform X(f) (t_0 = 0)	Character of the spectrum
A·rect(t/T) A for |t| < T/2	A, T	A·T·sinc(fT)	Real, even; zeros at f = k/T; X(0) = AT
A·tri(t/T) A(1 − |t|/T) for |t| < T	A, T	A·T·sinc²(fT)	Real, non-negative; zeros at f = k/T; X(0) = AT
A·e^(−t²/2σ²)	A, σ	A·σ√(2π)·e^(−2π²σ²f²)	Real Gaussian; width 1/(2πσ); X(0) = Aσ√(2π)
A·sinc(Bt)	A, B	(A/B)·rect(f/B)	Band-limited to |f| < B/2; X(0) = A/B
A·e^(−at)·u(t), a > 0	A, a	A / (a + j2πf)	Complex; |X| = A/√(a² + 4π²f²); phase −arctan(2πf/a)

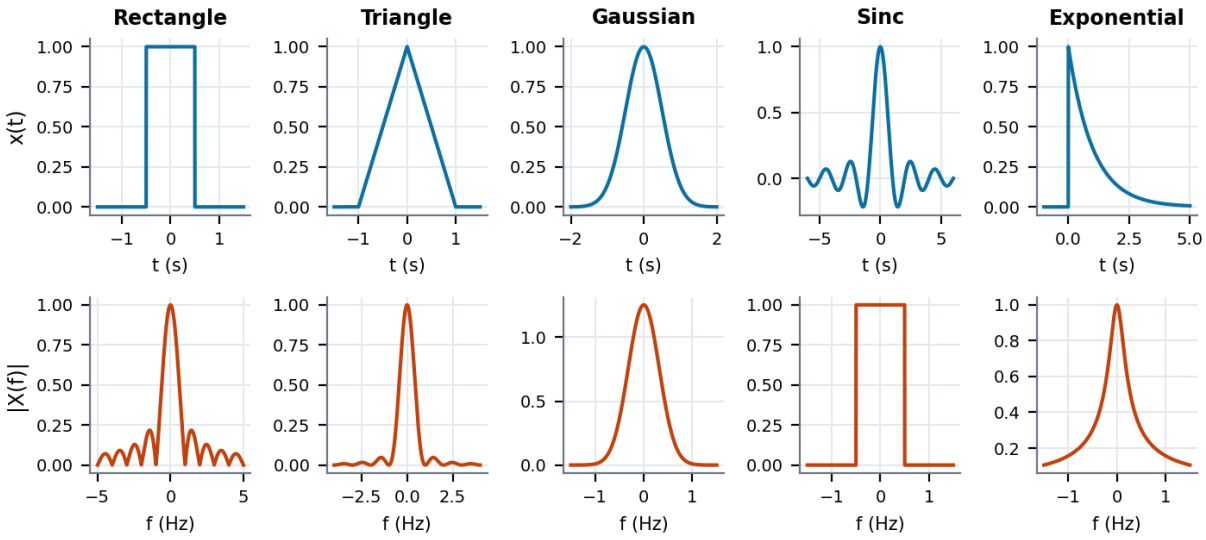


Figure 1. The five signals (top row) and the magnitudes of their transforms (bottom row) for A = 1 and the default widths.

3.1 Derivations

Rectangular pulse

X(f) = ∫ from −T/2 to T/2 of A e^(−j2πft) dt = A (e^(jπfT) − e^(−jπfT)) / (j2πf) = A sin(πfT)/(πf) = A·T·sinc(fT). The first zero occurs at f = 1/T, so a narrower pulse has a wider main lobe.

Triangular pulse

A triangle of half-width T and peak A is (A/T) times the convolution of two rectangles of width T. Convolution in time becomes multiplication in frequency, so X(f) = (A/T)·(T sinc(fT))² = A·T·sinc²(fT).

Gaussian pulse

Using $\int e^{(-\alpha t^2)} e^{(-j\omega t)}\, dt = \sqrt{(\pi/\alpha)}\, e^{(-\omega^2/4\alpha)}$ with $\alpha = 1/(2\sigma^2)$ and $\omega = 2\pi f$ gives $X(f) = A\,\sigma\,\sqrt{(2\pi)}\, e^{(-2\pi^2\sigma^2 f^2)}$. A Gaussian transforms to a Gaussian, and the product of the time width σ and the frequency width $1/(2\pi\sigma)$ is constant: the time-bandwidth trade-off.

Sinc pulse

From $\text{rect}(t/B) \leftrightarrow B\cdot\text{sinc}(fB)$, duality gives $B\cdot\text{sinc}(Bt) \leftrightarrow \text{rect}(f/B)$ (the rectangle is even). Dividing by B gives $\text{sinc}(Bt) \leftrightarrow (1/B)\,\text{rect}(f/B)$.

One-sided exponential

$X(f) = A \int$ from 0 to ∞ of $e^{(-(a + j2\pi f)t)}\, dt = A / (a + j2\pi f)$, valid for $a > 0$. Multiplying numerator and denominator by the conjugate gives $\text{Re } X = A\cdot a/(a^2 + 4\pi^2 f^2)$ and $\text{Im } X = -2\pi f\cdot A/(a^2 + 4\pi^2 f^2)$. The magnitude falls to $1/\sqrt{2}$ of its peak at $f = a/(2\pi)$.

4 System design and methodology

4.1 Requirements

- Real-time response: every control change must refresh all plots without perceptible delay.
- No installation: the tool should run from one file in any modern browser, on desktop or phone.
- Correctness first: plotted transforms are closed-form; numerical routines serve as independent checks.
- Readable in light and dark themes and usable on narrow screens.

4.2 Technology choice

The application is written in plain HTML, CSS and JavaScript with the HTML canvas element for drawing. No external library is needed, which keeps the file small, makes it work offline, and keeps every line of the mathematics visible for explanation in a viva.

4.3 Architecture

User input updates a small state object. A single function, render(), reads the state, evaluates the selected signal and its transform at a few hundred points, optionally runs the numerical engine, and hands arrays to a generic plotting function. Redraws are scheduled with requestAnimationFrame so that rapid slider movement produces at most one redraw per screen refresh.

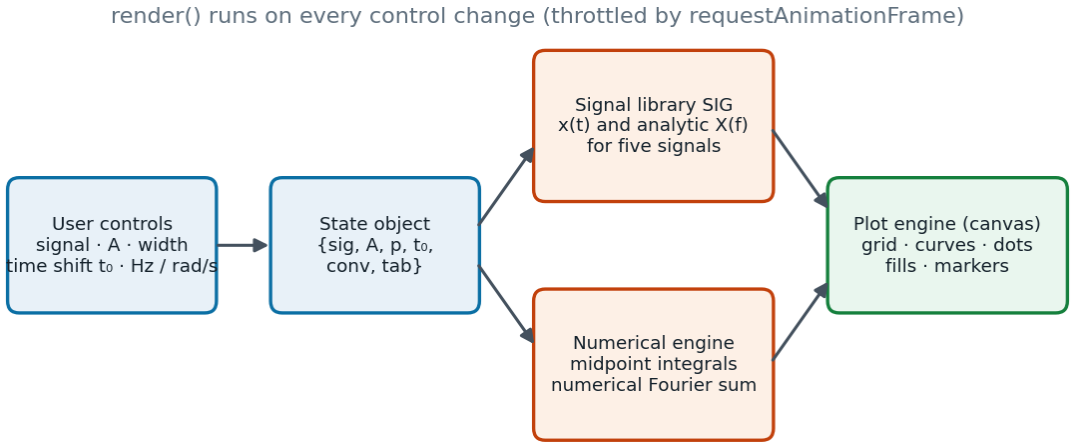


Figure 2. Block diagram of the application.

4.4 Handling of the frequency convention

Internally the transform is always evaluated in hertz. A constant c equals 1 in Hz mode and 2π in rad/s mode, and the axis variable v is converted by f = v/c. The inverse-area formula becomes x(0) = (1/c)∫X(v)dv and the duality constant becomes c, which reproduces both columns of the table in Section 2.1 with one piece of code.

4.5 Numerical methods

Table 3. Numerical methods and settings.

Task	Method	Settings
Curve plotting	Closed-form evaluation on a uniform grid	700 points per curve; window set per signal from its width plus $\lvert t_o \rvert$
Phase display	Principal value atan2(Im, Re), hidden where $\lvert X \rvert$ < 0.1 % of its peak	Avoids meaningless jumps where the spectrum is zero
Duality check	Numerical transform of the function X(t): Y(f) = Σ X(t□) e^(−j2πf t□) Δt	4001 samples over $\lvert t \rvert \le 40$ ($\lvert t \rvert \le 40 \cdot 2\pi$ in rad/s); 161 test frequencies
Duality verdict	RMS gap between numerical Y and predicted c·x(−f)	Pass if below 10 % of the predicted peak
Area checks	Midpoint sums of x(t) over $\lvert t \rvert \le 60$ s and of X over the equivalent frequency span	24 000 points each
Area verdict	Absolute error against closed-form X(0) and x(0)	Pass if below 2 % of max(|reference|, A)

A finite window truncates the slowly decaying tails of sinc-type functions, and abrupt edges cause Gibbs ripple. The tolerances above are chosen to be tight enough to catch a wrong formula yet loose enough to ignore these truncation effects.

5 Implementation of the graphical interface

The interface has a control panel and four tabs. The control panel is shared by all tabs so that a change made in one tab is visible in the others.

Table 4. Interface components.

Element	Purpose
Signal buttons	Select rectangle, triangle, Gaussian, sinc or exponential. Selecting a signal resets its width parameter to a sensible default.
Amplitude slider A	Scales the signal from 0.5 to 3.
Width slider	Adjusts T, σ, B or a depending on the signal; its label changes to match.
Time-shift slider t_0	Delays the signal from −2 s to +2 s.
Frequency-axis toggle	Switches between f in Hz and ω in rad/s; formulas, duality constant and area formula update.
Explorer tab	Five plots: x(t), |X|, ∠X, Re X, Im X, with the transform pair printed above them.
Duality tab	Left card: x(t) and X(f). Right card: X(t) and x(−f), with numerical dots over the predicted curve and an RMS-gap verdict.
Area tab	Shaded-area plots of x(t) and Re X, with two result panels comparing closed-form and numerical values.
Viva notes tab	Written derivation of the transform of a constant by duality, with consistency checks.

The complete source code is listed in Appendix C, and screenshots of the running program for every signal and every tab are in Appendix D.

6 Results and verification

6.1 Explorer observations

- **Width trade-off.** For every signal, narrowing the pulse widens the spectrum and vice versa. Figure 3 shows three Gaussians: as σ grows the pulse spreads while the spectrum becomes taller and narrower, approaching the impulse that represents a constant.
- **Time shift.** Moving the pulse changes only the phase and the real and imaginary parts. Figure 4 shows a rectangle delayed by 0.5 s: |X| is identical to the unshifted case, the phase is a straight line with slope $-2\pi t_0$ (wrapped to $\pm\pi$), and Re and Im oscillate as the rotating phasor sweeps through them.
- **Exponential spectrum.** Because the signal is not even, its transform is complex. The phase moves smoothly from 0 towards $-\pi/2$ and the magnitude drops to 70.7 % at $f = a/(2\pi)$.

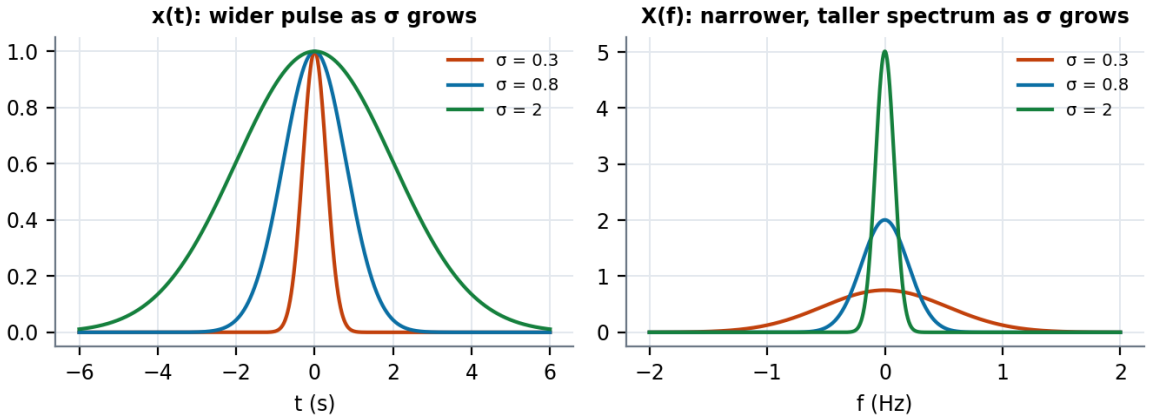


Figure 3. Gaussian pulses of increasing σ and their transforms.

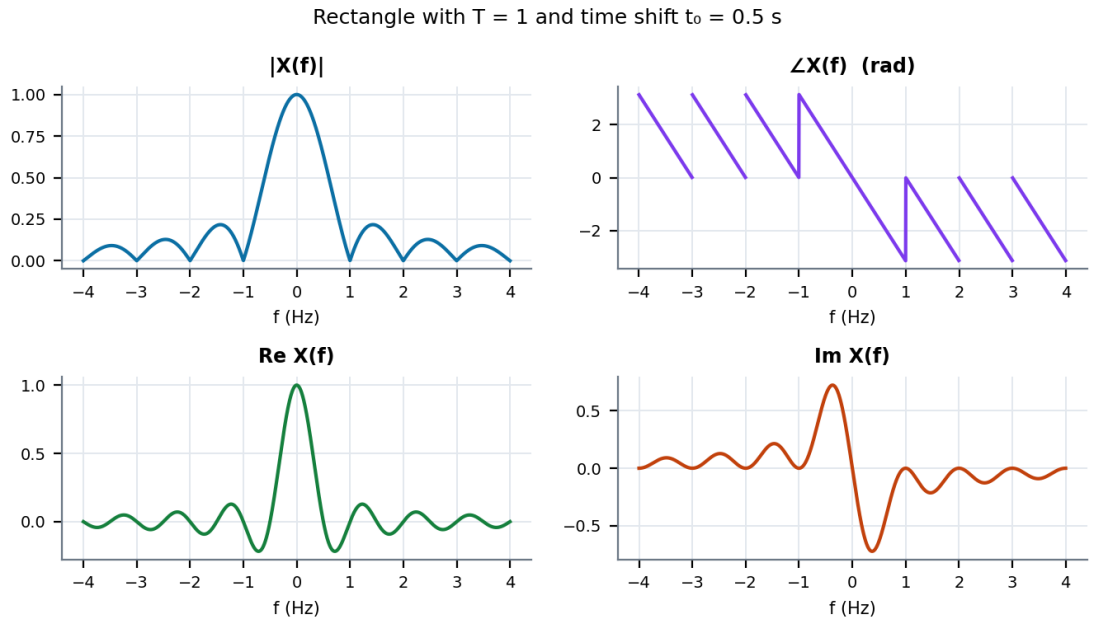


Figure 4. Magnitude, phase, real and imaginary parts for a rectangle delayed by $t_0 = 0.5$ s.

6.2 Duality verification

Figure 5 reproduces the Duality tab for the rectangle. The top row shows x(t) = rect(t) and X(f) = sinc(f). The bottom row shows X(t) = sinc(t) and its transform. The curve is the prediction x(−f) = rect(f); the dots come from numerically transforming the sinc over |t| ≤ 40. The dots follow the rectangle except for a small transition at the two edges, which is truncation of the sinc tail.

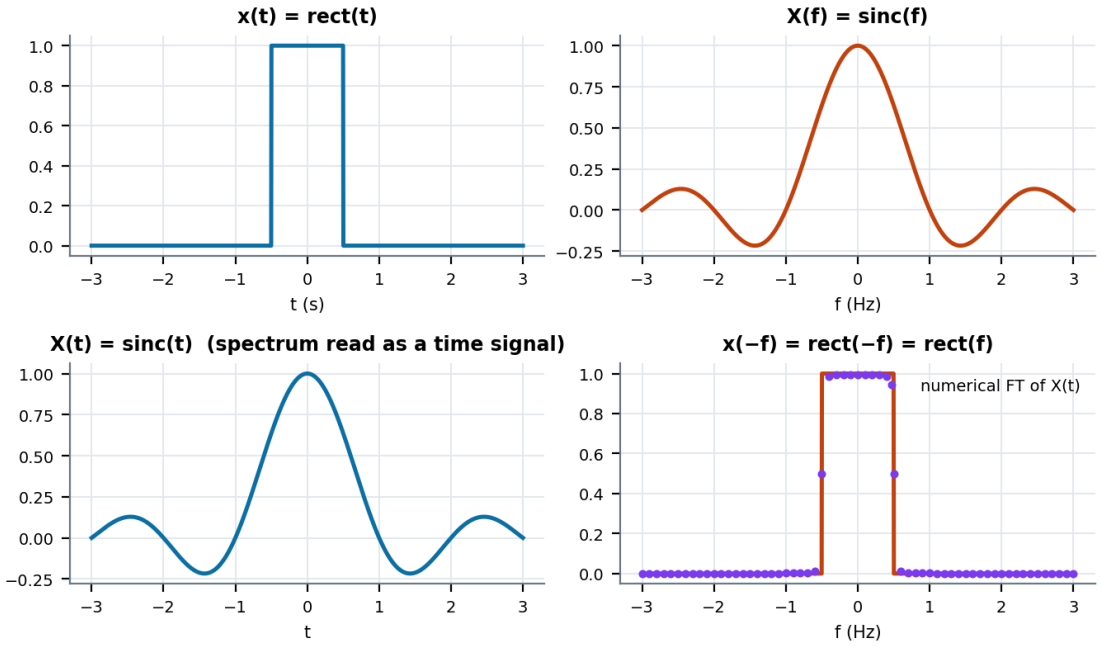


Figure 5. Duality for the rectangle: the transform of X(t) = sinc(t) is x(−f) = rect(f).

6.3 Area-property verification

Table 5 lists the closed-form and numerical values for the default parameters with $t_0 = 0$. The column ∫x dt should equal X(0), and the column ∫X df should equal x(0). Note that x(0) for the exponential is 0.5, the midpoint of the jump.

Table 5. Verification for A = 1, $t_0 = 0$ (Hz axis).

Signal	X(0) closed form	∫x(t)dt numerical	x(0) closed form	∫X(f)df numerical	Duality gap (% of peak)
Rectangle (T = 1)	1.0000	1.0000	1.0000	0.9966	1.76 %
Triangle (T = 1)	1.0000	1.0000	1.0000	0.9983	0.02 %
Gaussian (σ = 0.5)	1.2533	1.2533	1.0000	1.0000	0.00 %
Sinc (B = 1)	1.0000	0.9966	1.0000	1.0000	0.10 %
Exponential (a = 1)	1.0000	1.0000	0.5000	0.4992	0.21 %

Table 6 repeats the test with a delay $t_0 = 0.5$ s. Now the area under the spectrum must equal the shifted signal value at t = 0, which changes for each signal. For the rectangle the edge falls exactly at t = 0, so x(0) = 0.5; for the exponential the signal has not started, so x(0) = 0. The numerical sums reproduce each of these values.

Table 6. Verification for A = 1, t_0 = 0.5 s (Hz axis). X(0) does not depend on the shift, x(0) does.

Signal	X(0) closed form	∫x(t)dt numerical	x(0) closed form	∫X(f)df numerical	Duality gap (% of peak)
Rectangle (T = 1)	1.0000	1.0000	0.5000	0.4992	0.62 %
Triangle (T = 1)	1.0000	1.0000	0.5000	0.5000	0.02 %
Gaussian (σ = 0.5)	1.2533	1.2533	0.6065	0.6065	0.00 %
Sinc (B = 1)	1.0000	1.0000	0.6366	0.6366	0.11 %
Exponential (a = 1)	1.0000	1.0000	0.0000	0.0017	3.20 %

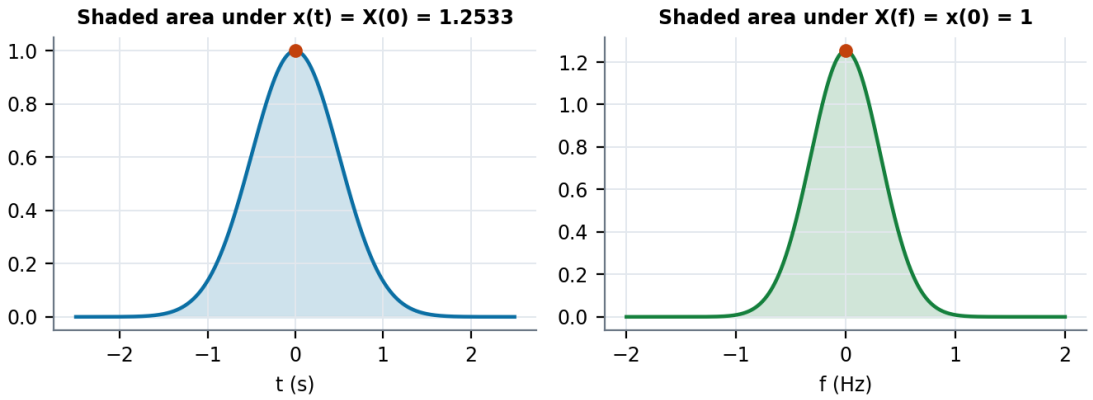


Figure 6. Area property for the Gaussian σ = 0.5: area under x(t) equals X(0), area under X(f) equals x(0).

6.4 Parameter sweep

To check robustness, all five signals were tested at their minimum, middle and maximum widths, with amplitudes 0.5 and 3, shifts of −2, 0, 1.3 and 2 s, and both axis conventions (240 cases). The largest area error was 0.7 % of the reference scale. The largest duality gap was 6.3 % of the peak, for the narrowest rectangle (T = 0.2): its transform X(t) = 0.2·sinc(0.2t) decays slowly, so the 80 s window cuts off many lobes and blurs the edges of the predicted rectangle.

6.5 Discussion

- The closed-form transforms and the independent numerical sums agree, which confirms the formulas, the shift implementation and the conversion factors between the two conventions.
- Remaining differences are explained by the finite window, not by the formulas: they shrink when the window is enlarged.
- The rad/s mode gives identical relative errors to the Hz mode, which shows that the 2π factors in duality and in the area formula are consistent.

7. Limitations and future work

- Only five signals are offered; a user-entered formula or a cosine and impulse train would extend the catalogue.
- The numerical checks use finite windows, so very slowly decaying functions show small truncation errors near edges.
- Phase is shown as the principal value in $(-\pi, \pi]$; an unwrapped-phase option would make group delay easier to read.
- The tool could add convolution and Parseval demonstrators, an option to export plots, and a screen-reader description of each plot.

8 Conclusion

The Continuous Fourier Transform Explorer meets the project brief: it provides five selectable signals with adjustable parameters, live magnitude, phase, real and imaginary plots, a side-by-side Duality demonstrator, and numerical verification of both halves of the Area Property. Closed-form results and independent numerical checks agree to within the limits set by finite integration windows, in both the Hz and rad/s conventions. The tool also supports viva preparation, since the transform of a constant, the role of the 2π factor and the effect of a time shift can all be seen directly.

References

1. A. V. Oppenheim, A. S. Willsky and S. H. Nawab, Signals and Systems, 2nd ed., Prentice Hall, 1997.
2. S. Haykin and B. Van Veen, Signals and Systems, 2nd ed., Wiley, 2002.
3. B. P. Lathi and R. A. Green, Linear Systems and Signals, 3rd ed., Oxford University Press, 2018.
4. R. N. Bracewell, The Fourier Transform and Its Applications, 3rd ed., McGraw-Hill, 2000.

Appendix A Formula sheet

Pair or property	Formula (Hz form)
Rectangle	$A \cdot \mathrm{rect}(t/T) \leftrightarrow A \cdot T \cdot \mathrm{sinc}(fT)$
Triangle	$A \cdot \mathrm{tri}(t/T) \leftrightarrow A \cdot T \cdot \mathrm{sinc}^2(fT)$
Gaussian	$A \cdot e^{(-t^2/2\sigma^2)} \leftrightarrow A \cdot \sigma\sqrt{(2\pi)} \cdot e^{(-2\pi^2\sigma^2 f^2)}$
Sinc	$A \cdot \mathrm{sinc}(Bt) \leftrightarrow (A/B) \cdot \mathrm{rect}(f/B)$
Exponential	$A \cdot e^{(-at)} \cdot u(t) \leftrightarrow A/(a + j2\pi f)$
Time shift	$x(t - t_o) \leftrightarrow X(f) \cdot e^{(-j2\pi f t_o)}$
Time scaling	$x(at) \leftrightarrow (1/\lvert a \rvert) \cdot X(f/a)$
Duality	$X(t) \leftrightarrow x(-f)$ (rad/s: $2\pi \cdot x(-\omega)$)
Area	$X(0) = \int x(t)dt$, $x(0) = \int X(f)df$
Impulse and constant	$\delta(t) \leftrightarrow 1$, $1 \leftrightarrow \delta(f)$

Appendix B Complete source code

The whole application is a single file, `fourier_explorer.html` (243 lines). To run it, save the listing as a file with the extension .html and open it in any modern browser; no installation, server or internet connection is needed. Line numbers are shown at the left and are not part of the code. Long lines wrap onto the next row.

B.1 Head and stylesheet (lines 1–42)

```
<!doctype html>
<html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Continuous Fourier Transform Explorer</title>
<style>
:root{--bg:#eef2f5;--card:#fff;--ink:#14212b;--mut:#5f6f7d;--line:#d3dce4;--grid:#e6ecf1;--c1:#0b6fa4;--c2:#c2410c;--c3:#15803d;--c4:#7c3aed;--acc:#0b6fa4;--ok:#15803d;--bad:#b91c1c;
box-sizing:border-box;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px);color-scheme:light}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#0e151b;--card:#16202a;--ink:#e4ebf1;--mut:#92a3b2;--line:#27343f;--grid:#1f2b35;--c1:#4cb4ec;--c2:#fb923c;--c3:#4ade80;--c4:#a78bfa;--acc:#1d86bd;--ok:#4ade80;--bad:#f87171;color-scheme:dark}}
:root[data-theme="dark"]{--bg:#0e151b;--card:#16202a;--ink:#e4ebf1;--mut:#92a3b2;--line:#27343f;--grid:#1f2b35;--c1:#4cb4ec;--c2:#fb923c;--c3:#4ade80;--c4:#a78bfa;--acc:#1d86bd;--ok:#4ade80;--bad:#f87171;color-scheme:dark}
html{scroll-padding-top:env(safe-area-inset-top,0px)}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--ink);font:15px/1.5 system-ui,-apple-system,"Segoe UI",Roboto,sans-serif}
main{max-width:1120px;margin:0 auto;padding:18px 14px 40px}
h1{font:600 1.5rem/1.2 Georgia,"Times New Roman",serif;margin:0 0 4px}
h3{margin:0 0 6px;font:600 1rem Georgia,serif}
.sub{color:var(--mut);margin:0 0 14px;font-size:.9rem}
.card{background:var(--card);border:1px solid var(--line);border-radius:10px;padding:12px 14px}
.controls{display:grid;gap:12px;margin-bottom:12px}
.row{display:flex;flex-wrap:wrap;align-items:center;gap:8px 12px}
.row>label{min-width:100px;color:var(--mut);font-size:.85rem}
.segs{display:flex;flex-wrap:wrap;gap:6px}
.segs button{border:1px solid var(--line);background:var(--bg);color:var(--ink);padding:6px 12px;border-radius:6px;font:inherit;font-size:.88rem;cursor:pointer}
.segs button.on{background:var(--acc);border-color:var(--acc);color:#fff}
.segs button:focus-visible,input:focus-visible{outline:2px solid var(--acc);outline-offset:2px}
.sliders{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:10px 22px}
.sliders label{display:flex;justify-content:space-between;font-size:.85rem;color:var(--mut)}
.sliders b{color:var(--ink);font-variant-numeric:tabular-nums}
input[type=range]{width:100%;accent-color:var(--acc)}
#tabs{margin:14px 0 12px}
.pane{display:none}.pane.on{display:block}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(310px,1fr));gap:12px}
.grid2{display:grid;grid-template-columns:repeat(auto-fit,minmax(340px,1fr));gap:12px}
.stack{display:grid;gap:10px}
canvas{width:100%;height:210px;display:block}
.eq{font-family:ui-monospace,Menlo,Consolas,monospace;font-size:.85rem;background:var(--bg);padding:8px 10px;border-radius:6px;margin:0 0 12px;overflow-x:auto;white-space:nowrap}
.note{color:var(--mut);font-size:.86rem;margin:8px 2px 0}
.res{display:grid;grid-template-columns:auto 1fr;gap:3px 14px;font-variant-numeric:tabular-nums;font-size:.92rem}
.res>span:nth-child(odd){color:var(--mut)}
.ok{color:var(--ok);font-weight:600}.bad{color:var(--bad);font-weight:600}
.viva p,.viva li{max-width:74ch}.viva ol{padding-left:20px}
.viva code{font-family:ui-monospace,Menlo,Consolas,monospace;background:var(--bg);padding:1px 6px;border-radius:4px}
</style></head><body>
```

B.2 Body: page structure (lines 43–110)

```
<main>
<h1>Continuous Fourier Transform Explorer</h1>
<p class="sub" id="conv-eq"></p>

<section class="card controls">
 <div class="row"><label>Signal</label><div class="segs" id="sigs"></div></div>
 <div class="sliders">
  <div><label>Amplitude A <b id="vA"></b></label><input type="range" id="sA" min="0.5" max="3" step="0.1"></div>
  <div><label><span id="lP">Width</span> <b id="vP"></b></label><input type="range" id="sP" step="0.05"></div>
```

```
  <div><label>Time shift t₀ (s) <b id="vT"></b></label><input type="range" id="sT" min="-2" max="2" step="0.05"></div>
 </div>
 <div class="row"><label>Frequency axis</label><div class="segs" id="conv"></div></div>
</section>

<nav class="segs" id="tabs"></nav>

<section class="pane" id="p-explore">
 <div class="eq" id="pairEq"></div>
 <div class="grid">
  <div class="card"><canvas id="c_x"></canvas></div>
  <div class="card"><canvas id="c_m"></canvas></div>
  <div class="card"><canvas id="c_p"></canvas></div>
  <div class="card"><canvas id="c_r"></canvas></div>
  <div class="card"><canvas id="c_i"></canvas></div>
 </div>
 <p class="note">Phase is hidden where |X| is below 0.1% of its peak, since it is undefined there. A time shift t₀ multiplies X by e^(−j2πft₀): magnitude is unchanged, phase gains a straight line, and real and imaginary parts mix.</p>
</section>

<section class="pane" id="p-dual">
 <div class="grid2">
  <div class="card stack"><h3 id="dL">x(t) ⇿ X(f)</h3><canvas id="d1"></canvas><canvas id="d2"></canvas></div>
  <div class="card stack"><h3 id="dR">X(t) ⇿ x(−f)</h3><canvas id="d3"></canvas><canvas id="d4"></canvas></div>
 </div>
 <div class="card" style="margin-top:12px"><div class="res" id="dualRes"></div>
 <p class="note" id="dualNote"></p></div>
</section>

<section class="pane" id="p-area">
 <div class="grid2">
  <div class="card"><canvas id="a1"></canvas></div>
  <div class="card"><canvas id="a2"></canvas></div>
 </div>
 <div class="grid2" style="margin-top:12px">
  <div class="card"><h3>X(0) = ∫ x(t) dt</h3><div class="res" id="ar1"></div></div>
  <div class="card"><h3 id="ar2t"></h3><div class="res" id="ar2"></div></div>
 </div>
 <p class="note">Numerical integrals use a midpoint sum over a wide finite window (|t| ≤ 60 s and the equivalent frequency span). At a jump discontinuity (for example the start of the exponential at t₀ = 0) the inverse transform returns the midpoint of the jump, which is why x(0) = A/2 there.</p>
</section>

<section class="pane" id="p-viva"><div class="card viva">
 <h3>Viva: what is the Fourier transform of x(t) = 1?</h3>
 <p><b>Answer:</b> <code>1 ⇿ δ(f)</code> in the Hz form, which is the same as <code>1 ⇿ 2π δ(ω)</code> in the rad/s form.</p>
 <p><b>Derivation using duality, starting from δ(t) ⇿ 1</b></p>
 <ol>
  <li>Take x(t) = δ(t). By the sifting property, X(f) = ∫ δ(t) e^(−j2πft) dt = e⁰ = 1. So δ(t) ⇿ 1.</li>
  <li>Duality says that if x(t) ⇿ X(f), then X(t) ⇿ x(−f). Here X(f) = 1 for all f, so X(t) is the constant 1.</li>
  <li>Therefore 1 ⇿ x(−f) = δ(−f) = δ(f), because δ is an even function.</li>
  <li>In angular frequency, δ(f) = δ(ω/2π) = 2π δ(ω), so 1 ⇿ 2π δ(ω). The extra 2π is the same factor the Duality tab shows in rad/s mode: X(t) ⇿ 2π x(−ω).</li>
 </ol>
 <p><b>Consistency checks</b></p>
 <ul>
  <li>Area property: X(0) = ∫ 1 dt = ∞, which matches the infinite spike δ(0). Also ∫ δ(f) df = 1 = x(0).</li>
  <li>Try it: pick Gaussian and raise σ. The pulse flattens toward a constant, the spectrum narrows toward a spike, and the area tab still reports ∫X df = x(0) = A.</li>
  <li>Reflection is visible with the exponential: its spectrum A/(a + j2πf) used as a time signal transforms to A·e^(af)·u(−f), a mirrored exponential, which is the x(−f) in the duality statement.</li>
 </ul>
</div></section>
</main>

```

B.3 JavaScript: mathematics, plotting and interface (lines 111–243)

```
<script>
const PI=Math.PI,TAU=2*PI,$=id=>document.getElementById(id);
const sinc=x=>Math.abs(x)<1e-12?1:Math.sin(PI*x)/(PI*x);
const SIG={
 rect:{n:'Rectangle',pl:'Width T',min:.2,max:4,def:1,
  x:(t,A,p)=>{const a=Math.abs(t);return a<p/2-1e-9?A:a<p/2+1e-9?A/2:0},
  X:(f,A,p)=>[A*p*sinc(f*p),0],X0:(A,p)=>A*p,
  f:['A·rect((t−t₀)/T)','A·T·sinc(fT)·e^(−j2πft₀)']},
 tri:{n:'Triangle',pl:'Half-width T',min:.2,max:3,def:1,
  x:(t,A,p)=>{const a=Math.abs(t);return a<p?A*(1-a/p):0},
```

```
  X:(f,A,p)=>[A*p*sinc(f*p)**2,0],X0:(A,p)=>A*p,
  f:['A·tri((t−t₀)/T)','A·T·sinc²(fT)·e^(−j2πft₀)']},
 gauss:{n:'Gaussian',pl:'Spread σ',min:.1,max:3,def:.5,
  x:(t,A,p)=>A*Math.exp(-t*t/(2*p*p)),
  X:(f,A,p)=>[A*p*Math.sqrt(TAU)*Math.exp(-2*PI*PI*p*p*f*f),0],X0:(A,p)=>A*p*Math.sqrt(TAU),
  f:['A·e^(−(t−t₀)²/2σ²)','A·σ√(2π)·e^(−2π²σ²f²)·e^(−j2πft₀)']},
 sinc:{n:'Sinc',pl:'Bandwidth B',min:.5,max:4,def:1,
  x:(t,A,p)=>A*sinc(p*t),
  X:(f,A,p)=>{const a=Math.abs(f);return[a<p/2-1e-9?A/p:a<p/2+1e-9?A/(2*p):0,0]},X0:(A,p)=>A/p,
  f:['A·sinc(B(t−t₀))','(A/B)·rect(f/B)·e^(−j2πft₀)']},
 exp:{n:'Exponential',pl:'Decay a',min:.3,max:5,def:1,
  x:(t,A,p)=>t>1e-9?A*Math.exp(-p*t):t>-1e-9?A/2:0,
  X:(f,A,p)=>{const d=p*p+TAU*TAU*f*f;return[A*p/d,-A*TAU*f/d]},X0:(A,p)=>A/p,
  f:['A·e^(−a(t−t₀))·u(t−t₀)','A/(a+j2πf)·e^(−j2πft₀)']}
};
const st={sig:'rect',A:1,p:1,t0:0,conv:'hz',tab:'explore'};

/* ---------- plotting ---------- */
function cv(id){const c=$(id),r=c.getBoundingClientRect(),d=devicePixelRatio||1;c.width=Math.round(r.width*d);c.height=Math.round(r.height*d);const g=c.getContext('2d');g.setTransform(d,0,0,d,0,0);return[g,r.width,r.height]}
function nice(a,b){const raw=(b-a)/5,p=Math.pow(10,Math.floor(Math.log10(raw))),m=raw/p,s=(m<1.5?1:m<3?2:m<7?5:10)*p,o=[];for(let v=Math.ceil(a/s)*s;v<=b+1e-9;v+=s)o.push(+v.toFixed(10));return o}
const fmt=v=>Math.abs(v)<1e-9?'0':(Math.abs(v)>=1000||Math.abs(v)<.01)?v.toExponential(0):String(+v.toPrecision(3));
function plot(id,xs,ser,o={}){
 const[g,W,H]=cv(id);if(!W)return;
 const css=getComputedStyle(document.documentElement),col=n=>css.getPropertyValue(n).trim();
 const L=46,R=10,T=26,B=28,pw=W-L-R,ph=H-T-B;
 let lo=1e99,hi=-1e99;

 ser.forEach(q=>{q.y.forEach(v=>{if(isFinite(v)){lo=Math.min(lo,v);hi=Math.max(hi,v)}});if(q.fill){lo=Math.min(lo,0);hi=Math.max(hi,0)}});
 if(o.yr){lo=o.yr[0];hi=o.yr[1]}else{const d=(hi-lo)||1;lo-=.08*d;hi+=.08*d}
 const x0=xs[0],x1=xs[xs.length-1],X=v=>L+(v-x0)/(x1-x0)*pw,Y=v=>T+(hi-v)/(hi-lo)*ph;
 g.font='11px system-ui,sans-serif';g.lineWidth=1;
 g.fillStyle=col('--mut');g.textAlign='center';
 nice(x0,x1).forEach(v=>{g.strokeStyle=col('--grid');g.beginPath();g.moveTo(X(v),T);g.lineTo(X(v),T+ph);g.stroke();g.fillText(fmt(v),X(v),H-14)});
 g.textAlign='right';
 nice(lo,hi).forEach(v=>{g.strokeStyle=col('--grid');g.beginPath();g.moveTo(L,Y(v));g.lineTo(L+pw,Y(v));g.stroke();g.fillText(fmt(v),L-5,Y(v)+4)});
 g.strokeStyle=col('--mut');g.globalAlpha=.6;g.beginPath();g.moveTo(X(0),T);g.lineTo(X(0),T+ph);g.moveTo(L,Y(0));g.lineTo(L+pw,Y(0));g.stroke();g.globalAlpha=1;
 g.textAlign='center';g.fillStyle=col('--mut');g.fillText(o.xl||'',L+pw/2,H-2);
 g.save();g.beginPath();g.rect(L,T,pw,ph);g.clip();
 ser.forEach(q=>{const sx=q.x||xs,c=col(q.c);g.strokeStyle=c;g.fillStyle=c;g.lineWidth=q.w||1.8;

 if(q.dots){sx.forEach((x,i)=>{if(isFinite(q.y[i])){g.beginPath();g.arc(X(x),Y(q.y[i]),2.3,0,TAU);g.fill()}})}
  else{g.beginPath();let pen=false;sx.forEach((x,i)=>{const v=q.y[i];if(!isFinite(v)){pen=false;return}pen?g.lineTo(X(x),Y(v)):g.moveTo(X(x),Y(v));pen=true});g.stroke();
   if(q.fill){g.lineTo(X(sx[sx.length-1]),Y(0));g.lineTo(X(sx[0]),Y(0));g.globalAlpha=.18;g.fill();g.globalAlpha=1}}});
 (o.marks||[]).forEach(m=>{g.fillStyle=col(m.c);g.beginPath();g.arc(X(m.x),Y(m.y),4.5,0,TAU);g.fill()});
 g.restore();
 g.textAlign='left';g.fillStyle=col('--ink');g.font='600 12px system-ui,sans-serif';g.fillText(o.t||'',L,15);
 g.font='11px system-ui,sans-serif';g.textAlign='right';let rx=W-R;
 ser.slice().reverse().forEach(q=>{if(!q.l)return;g.fillStyle=col(q.c);g.fillText(q.l,rx,15);rx-=g.measureText(q.l).width+12});
}
const grid=(a,b,n)=>Array.from({length:n},(_,i)=>a+(b-a)*i/(n-1));

/* ---------- model ---------- */
function ranges(){const p=st.p;let a,b;
 switch(st.sig){case'rect':a=.9*p;b=Math.max(4/p,1.5);break;case'tri':a=1.3*p;b=Math.max(3/p,1.5);break;

 case'gauss':a=3.5*p;b=Math.max(1.3/p,.6);break;case'sinc':a=6/p;b=Math.max(p,1);break;default:a=Math.max(5/p,1.5);b=Math.max(.95*p,1.5)}
 return[a+Math.abs(st.t0)+.2,b]}

function render(){
 const hz=st.conv==='hz',c=hz?1:TAU,[tR,fR]=ranges(),nR=fR*c,S=SIG[st.sig],{A,p,t0}=st;
 const nu=hz?'f':'ω',un=hz?'Hz':'rad/s',k=c; // duality constant
 const xf=t=>S.x(t-t0,A,p);
 const Xn=v=>{const f=v/c,[a,b]=S.X(f,A,p),ph=-TAU*f*t0,cs=Math.cos(ph),sn=Math.sin(ph);return[a*cs-b*sn,a*sn+b*cs]};
 const mag=V=>V.map(v=>Math.hypot(v[0],v[1]));
 const Xlab='X('+nu+')';
 $('conv-eq').textContent=hz?'X(f) = ∫ x(t) e^(−j2πft) dt     x(t) = ∫ X(f) e^(j2πft) df':'X(ω) = ∫ x(t) e^(−jωt) dt     x(t) = (1/2π) ∫ X(ω) e^(jωt) dω';
 if(st.tab==='explore'){
```

```
  $('pairEq').textContent='x(t) = '+S.f[0]+'   ⇔   X(f) = '+S.f[1]+(hz?'':'   (X(ω) = X(f) at f = ω/2π)');
  const ts=grid(-tR,tR,700),ns=grid(-nR,nR,700),V=ns.map(Xn),m=mag(V),mx=Math.max(...m);
  plot('c_x',ts,[{y:ts.map(xf),c:'--c1',l:'real'}],{t:'x(t)',xl:'t (s)'});
  plot('c_m',ns,[{y:m,c:'--c1'}],{t:'|'+Xlab+'|',xl:nu+' ('+un+')'});
  plot('c_p',ns,[{y:V.map((v,i)=>m[i]>1e-3*mx?Math.atan2(v[1],v[0]):NaN),c:'--c4'}],{t:'∠'+Xlab+' (rad)',xl:nu+' ('+un+')',yr:[-3.6,3.6]});
  plot('c_r',ns,[{y:V.map(v=>v[0]),c:'--c3'}],{t:'Re '+Xlab,xl:nu+' ('+un+')'});
  plot('c_i',ns,[{y:V.map(v=>v[1]),c:'--c2'}],{t:'Im '+Xlab,xl:nu+' ('+un+')'});
 }
 if(st.tab==='dual'){
  $('dL').textContent='x(t) ⇔ '+Xlab;
  $('dR').textContent=hz?'X(t) ⇔ x(−f)':'X(t) ⇔ 2π·x(−ω)';
  const ts=grid(-tR,tR,700),ns=grid(-nR,nR,700),V=ns.map(Xn);
  plot('d1',ts,[{y:ts.map(xf),c:'--c1'}],{t:'x(t)',xl:'t (s)'});
  plot('d2',ns,[{y:mag(V),c:'--c1',l:'|X|'},{y:V.map(v=>v[0]),c:'--c3',l:'Re',w:1.2},{y:V.map(v=>v[1]),c:'--c2',l:'Im',w:1.2}],{t:Xlab,xl:nu+' ('+un+')'});
  const tt=grid(-nR,nR,700),Yt=tt.map(Xn);
  plot('d3',tt,[{y:mag(Yt),c:'--c1',l:'|X(t)|'},{y:Yt.map(v=>v[0]),c:'--c3',l:'Re',w:1.2},{y:Yt.map(v=>v[1]),c:'--c2',l:'Im',w:1.2}],{t:'X(t): the spectrum read as a time signal',xl:'t'});
  // numerical FT of y(t)=X(t)
  const Wi=40*c,N=4001,dt=2*Wi/(N-1),wf=TAU/c,yr=new Float64Array(N),yi=new Float64Array(N),tg=new Float64Array(N);
  for(let i=0;i<N;i++){tg[i]=-Wi+i*dt;const v=Xn(tg[i]);yr[i]=v[0];yi[i]=v[1]}
  const vn=grid(-tR,tR,161),nr=[],ni=[];let se=0;
  vn.forEach(v=>{const w=v*wf;let a=0,b=0;for(let i=0;i<N;i++){const cs=Math.cos(w*tg[i]),sn=Math.sin(w*tg[i]);a+=yr[i]*cs+yi[i]*sn;b+=yi[i]*cs-yr[i]*sn}
    a*=dt;b*=dt;nr.push(a);ni.push(b);const e=a-k*xf(-v);se+=e*e+b*b});
  const rms=Math.sqrt(se/vn.length),vs=grid(-tR,tR,700);
  plot('d4',vs,[{y:vs.map(v=>k*xf(-v)),c:'--c1',l:hz?'x(−f) predicted':'2π·x(−ω) predicted'},{x:vn,y:nr,dots:true,c:'--c2',l:'numerical FT of X(t)'}],{t:hz?'x(−f)':'2π·x(−ω)',xl:nu+' ('+un+')'});
  const pk=Math.max(1e-9,...vs.map(v=>Math.abs(k*xf(-v))));
  $('dualRes').innerHTML='<span>Pair 1</span><span>x(t) ⇔ '+Xlab+'</span><span>Pair 2</span><span>'+(hz?'X(t) ⇔ x(−f)':'X(t) ⇔ 2π·x(−ω)')+'</span><span>RMS gap, numerical vs predicted</span><span>'+rms.toExponential(2)+' ('+(100*rms/pk).toFixed(2)+'% of peak) <span class="'+(rms/pk<.1?'ok':'bad')+'">'+(rms/pk<.1?'duality verified':'check')+'</span></span>';
  $('dualNote').textContent='The red dots come from numerically transforming the blue spectrum X(t) over |t| ≤ '+Wi.toFixed(0)+'. They land on the predicted curve, so the transform of X(t) really is x(−f) (times 2π in rad/s). Any small ripple near jumps is truncation (Gibbs), not a failure of the property.';
 }
 if(st.tab==='area'){
  const ts=grid(-tR,tR,700),ns=grid(-nR,nR,700),V=ns.map(Xn),x0v=xf(0),X0=S.X0(A,p);
  plot('a1',ts,[{y:ts.map(xf),c:'--c1',fill:true}],{t:'x(t): shaded area = X(0)',xl:'t (s)',marks:[{x:0,y:x0v,c:'--c2'}]});
  plot('a2',ns,[{y:V.map(v=>v[0]),c:'--c3',fill:true}],{t:'Re '+Xlab+': shaded area'+(hz?'':'/2π')+' = x(0)',xl:nu+' ('+un+')',marks:[{x:0,y:X0,c:'--c2'}]});
  const n=24000,dt=120/n;let I1=0;for(let i=0;i<n;i++)I1+=xf(-60+(i+.5)*dt);I1*=dt;
  const Fi=60*c,dn=2*Fi/n;let Ir=0,Ii=0;for(let i=0;i<n;i++){const v=Xn(-Fi+(i+.5)*dn);Ir+=v[0];Ii+=v[1]}Ir*=dn/c;Ii*=dn/c;
  const tol=.02*Math.max(Math.abs(X0),A),tol2=.02*Math.max(Math.abs(x0v),A),f4=v=>v.toFixed(4);
  const e1=Math.abs(I1-X0),e2=Math.hypot(Ir-x0v,Ii);
  $('ar1').innerHTML='<span>Analytic X(0)</span><span>'+f4(X0)+'</span><span>Numerical ∫x(t)dt</span><span>'+f4(I1)+'</span><span>Error</span><span>'+e1.toExponential(2)+' <span class="'+(e1<tol?'ok':'bad')+'">'+(e1<tol?'verified':'check')+'</span></span>';
  $('ar2t').textContent=hz?'x(0) = ∫ X(f) df':'x(0) = (1/2π) ∫ X(ω) dω';
  $('ar2').innerHTML='<span>Analytic x(0)</span><span>'+f4(x0v)+'</span><span>Numerical real part</span><span>'+f4(Ir)+'</span><span>Numerical imaginary part</span><span>'+f4(Ii)+'</span><span>Error</span><span>'+e2.toExponential(2)+' <span class="'+(e2<tol2?'ok':'bad')+'">'+(e2<tol2?'verified':'check')+'</span></span>';
 }
}

/* ---------- UI ---------- */
let q=0;const draw=()=>{if(!q)q=requestAnimationFrame(()=>{q=0;render()})};
function segs(id,items,get,set){const el=$(id);el.innerHTML='';items.forEach(([k,l])=>{const b=document.createElement('button');b.textContent=l;b.className=get()===k?'on':'';b.onclick=()=>{set(k);ui();draw()};el.appendChild(b)})}
function sliders(){const S=SIG[st.sig];$('lP').textContent=S.pl;const sp=$('sP');sp.min=S.min;sp.max=S.max;sp.value=st.p;$('sA').value=st.A;$('sT').value=st.t0;lab()}
function lab(){$('vA').textContent=(+st.A).toFixed(1);$('vP').textContent=(+st.p).toFixed(2);$('vT').textContent=(+st.t0).toFixed(2)}
function ui(){
 segs('sigs',Object.keys(SIG).map(k=>[k,SIG[k].n]),()=>st.sig,k=>{st.sig=k;st.p=SIG[k].def;sliders()});
 segs('conv',[['hz','f in Hz'],['rad','ω in rad/s']],()=>st.conv,k=>st.conv=k);
 segs('tabs',[['explore','Explorer'],['dual','Duality'],['area','Area property'],['viva','Viva notes']],()=>st.tab,k=>st.tab=k);
 ['explore','dual','area','viva'].forEach(k=>$('p-'+k).classList.toggle('on',st.tab===k));
}

[['sA','A'],['sP','p'],['sT','t0']].forEach(([id,key])=>$(id).addEventListener('input',e=>{st[key]=parseFloat(e.target.value);lab();draw()}));
addEventListener('resize',draw);
matchMedia('(prefers-color-scheme: dark)').addEventListener('change',draw);
st.p=SIG[st.sig].def;sliders();ui();render();
```

```
</script>
</body></html>
```

Appendix C Program output (screenshots)

The screenshots below were captured from the running program in a desktop browser window (1120 px wide, light theme). The page was loaded and operated through all four tabs, all five signals and both axis modes, and the browser reported no JavaScript errors. Unless stated, the amplitude is A = 1 and the time shift is $t_0 = 0$.

C.1 Explorer tab: the five signals

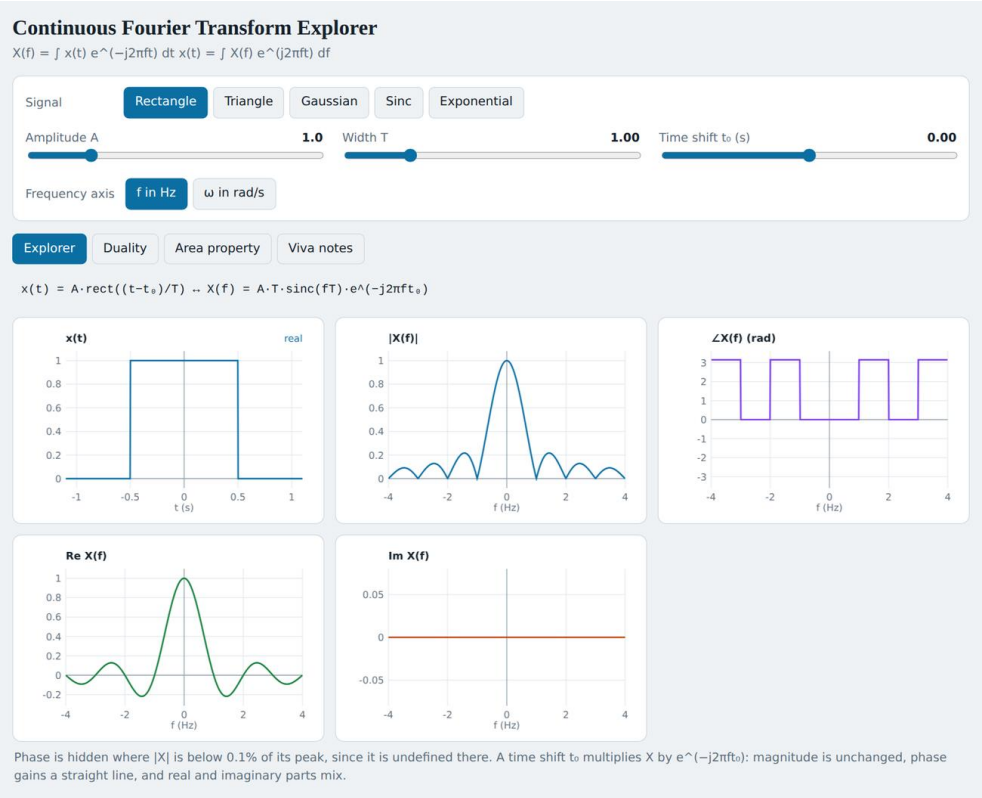


Figure 7. Explorer, rectangle (T = 1). The magnitude is a sinc with zeros at multiples of 1 Hz; the phase switches between 0 and π where the sinc changes sign.

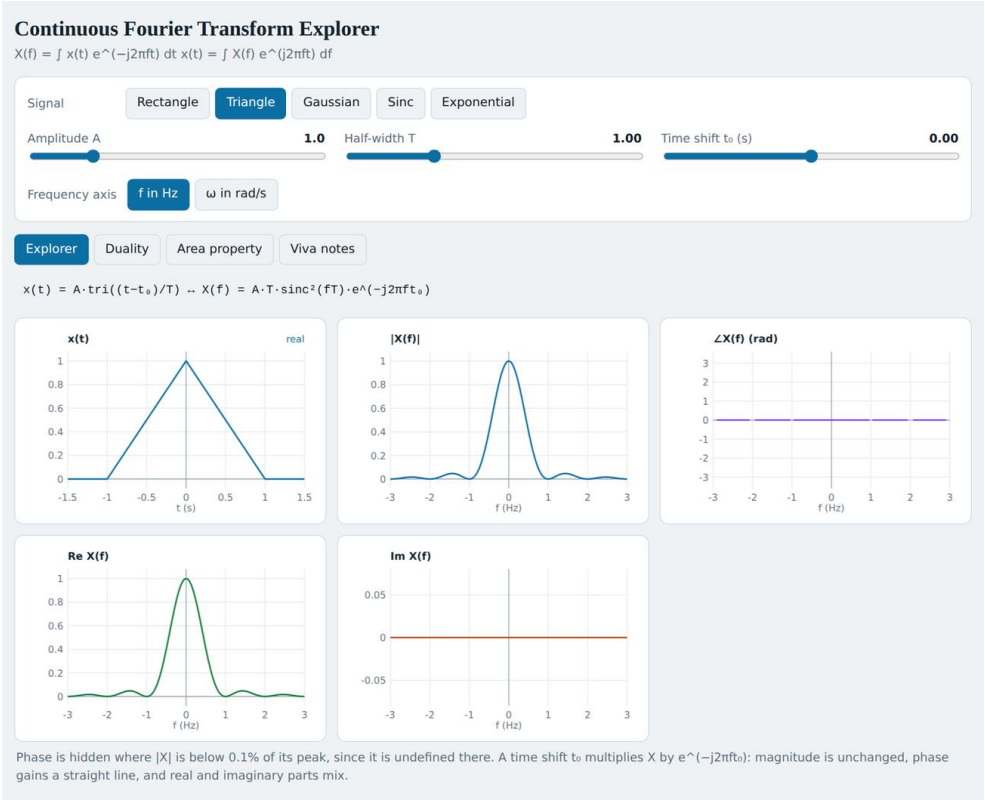


Figure 8. Explorer, triangle (T = 1). The spectrum is sinc², real and non-negative, so the phase is 0 wherever it is shown.

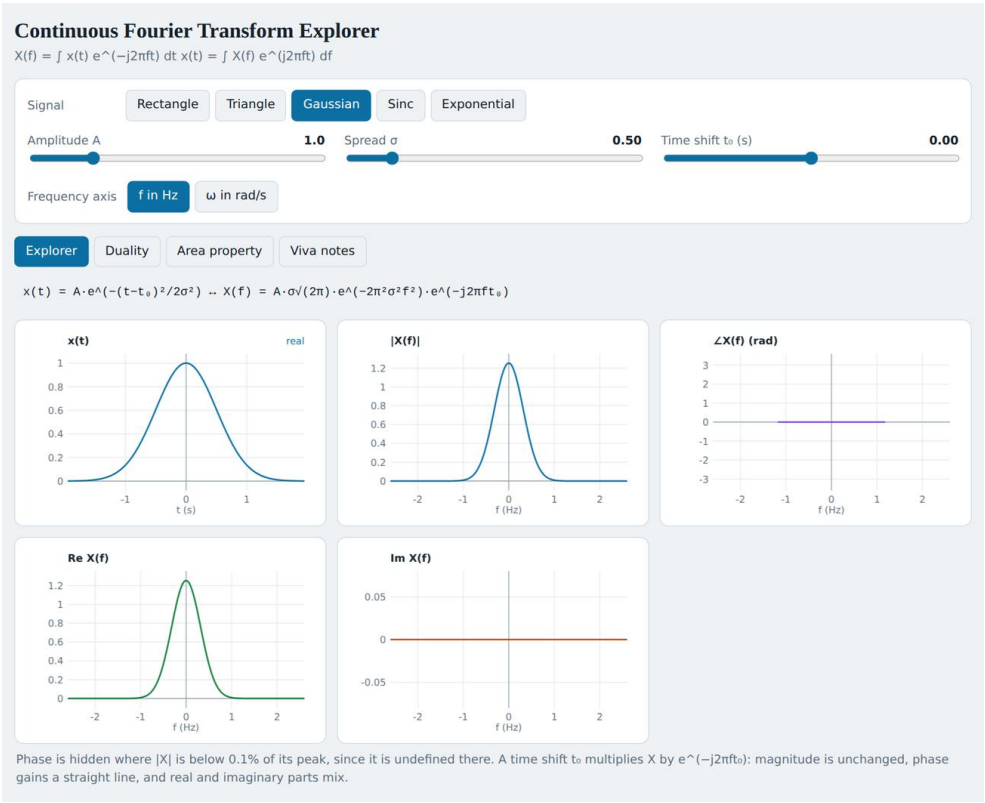


Figure 9. Explorer, Gaussian (σ = 0.5). The transform is also a Gaussian, real and positive, with no side lobes.

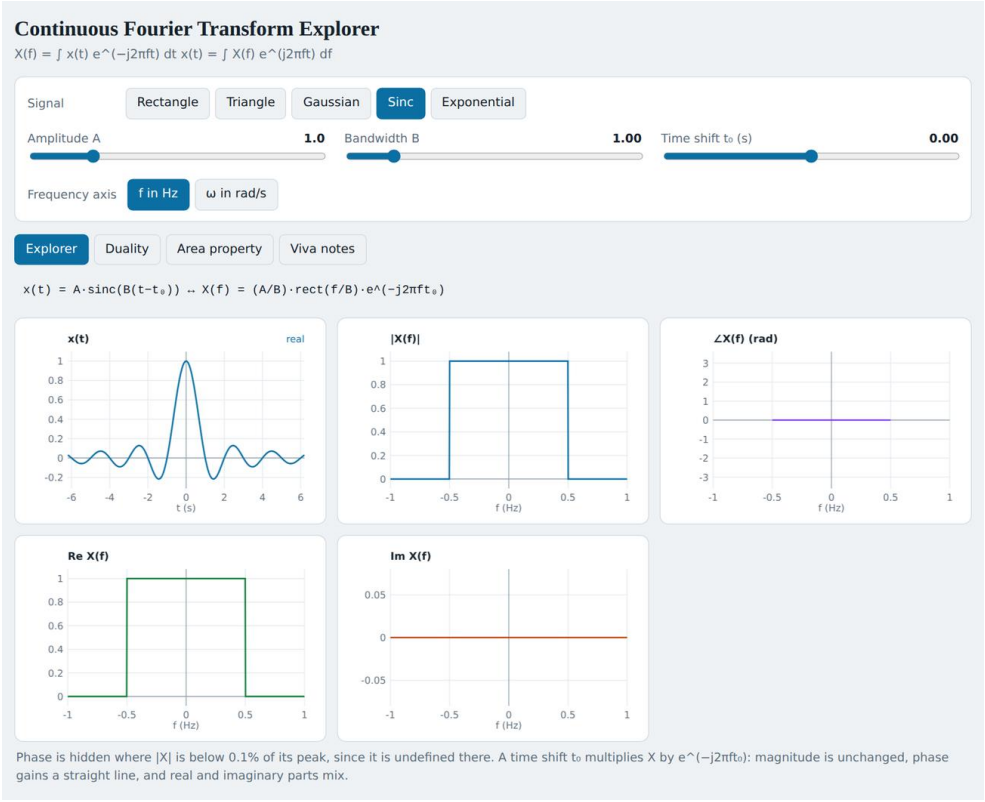


Figure 10. Explorer, sinc (B = 1). The spectrum is a rectangle limited to |f| < 0.5 Hz; the phase is hidden outside the band, where |X| = 0.

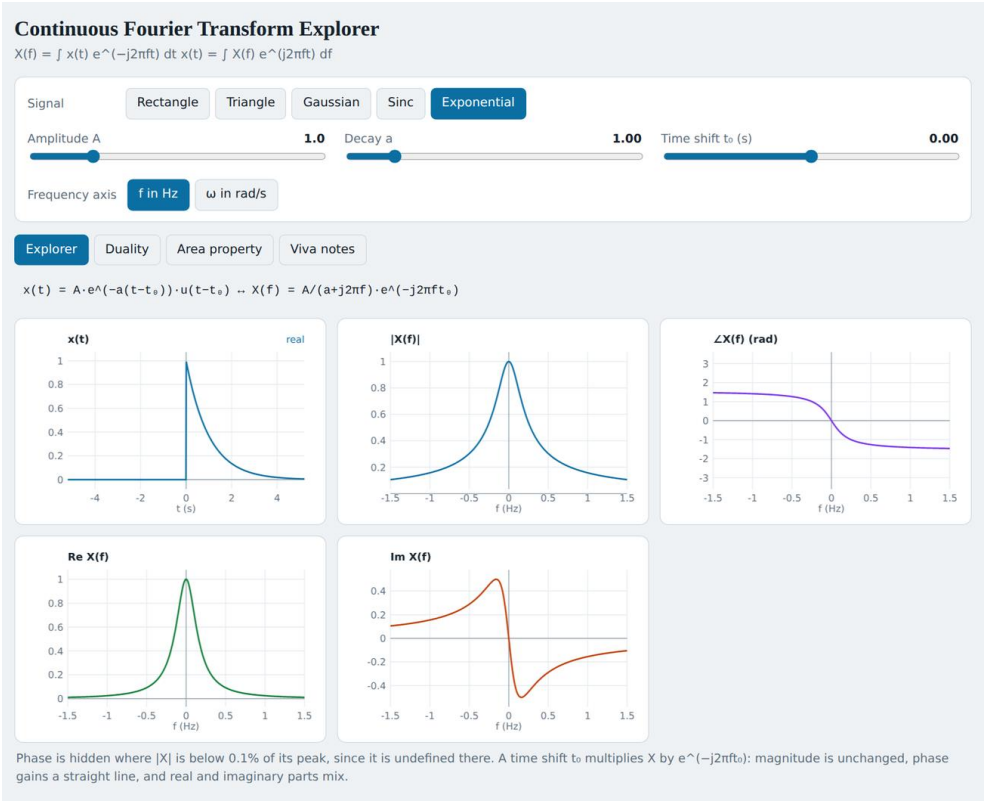


Figure 11. Explorer, exponential (a = 1). The spectrum is complex: Re X is even, Im X is odd, and the phase falls smoothly towards −π/2 for large positive f.

C.2 Explorer tab: effect of a time shift

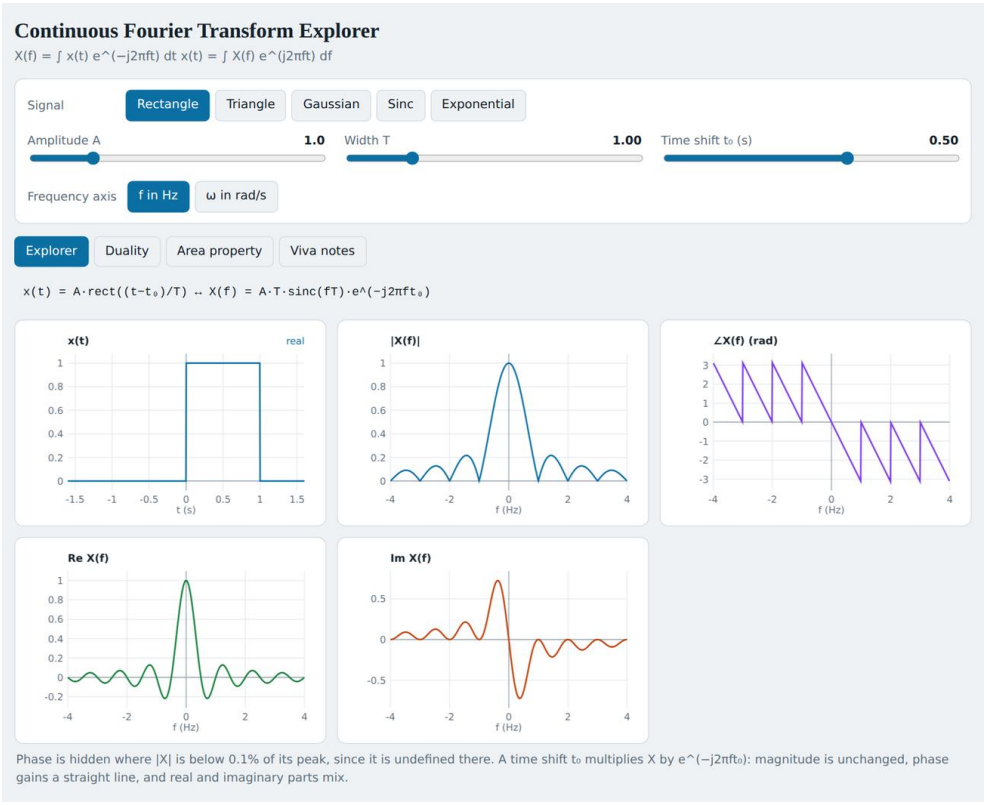


Figure 12. Explorer, rectangle (T = 1) delayed by t_0 = 0.5 s. The magnitude is unchanged; the phase becomes a wrapped straight line and Re X and Im X oscillate.

C.3 Duality tab

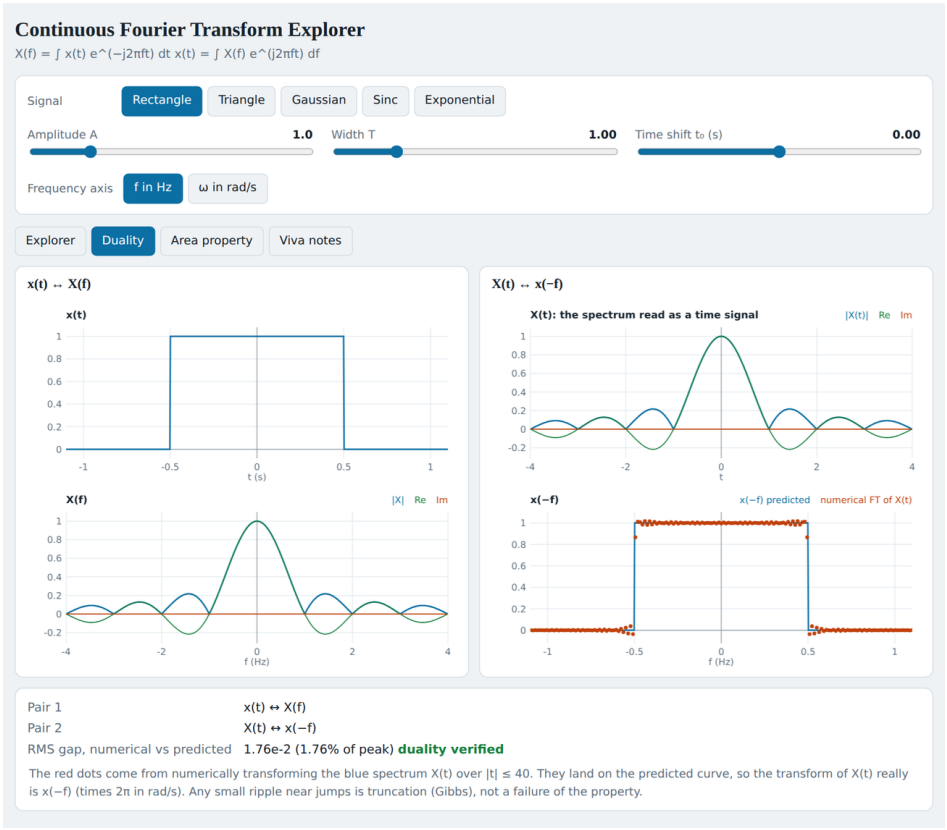


Figure 13. Duality, rectangle, Hz axis. Left: x(t) ↔ X(f), rect ↔ sinc. Right: the sinc read as a time signal transforms to x(−f) = rect(f); the red dots are the numerical transform. Result: RMS gap 1.76 % of peak, "duality verified".

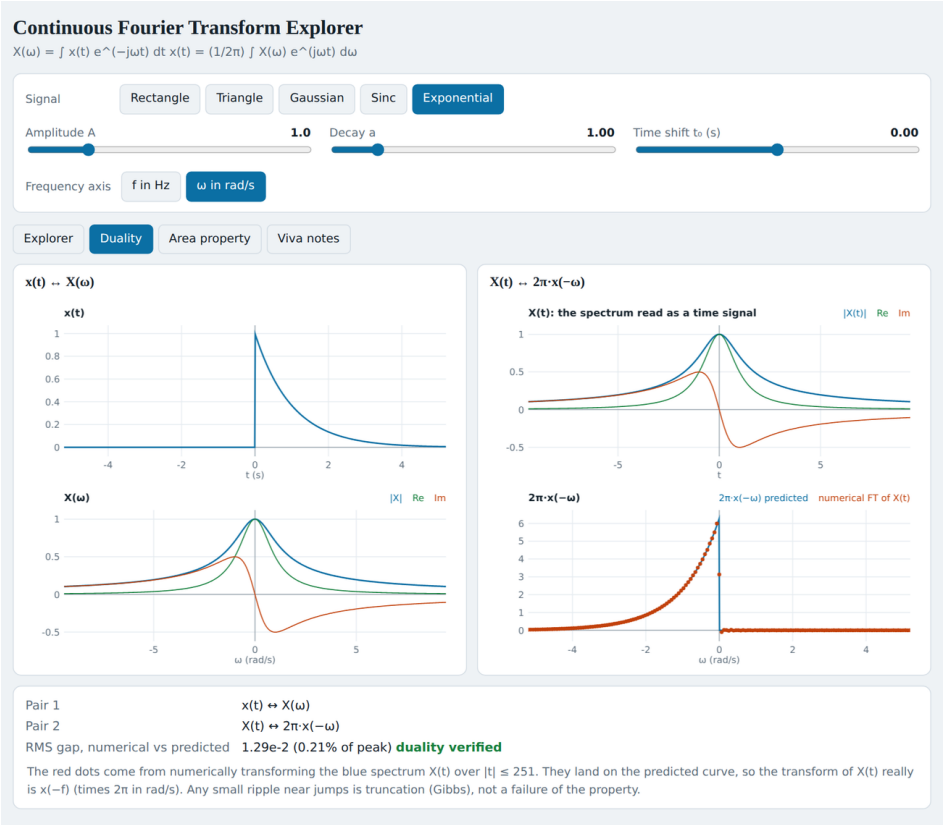


Figure 14. Duality, exponential (a = 1), ω axis. The right-hand pair reads X(t) ↔ 2π·x(−ω); the predicted curve is the mirrored exponential. Result: RMS gap 0.21 % of peak, "duality verified".

C.4 Area property tab

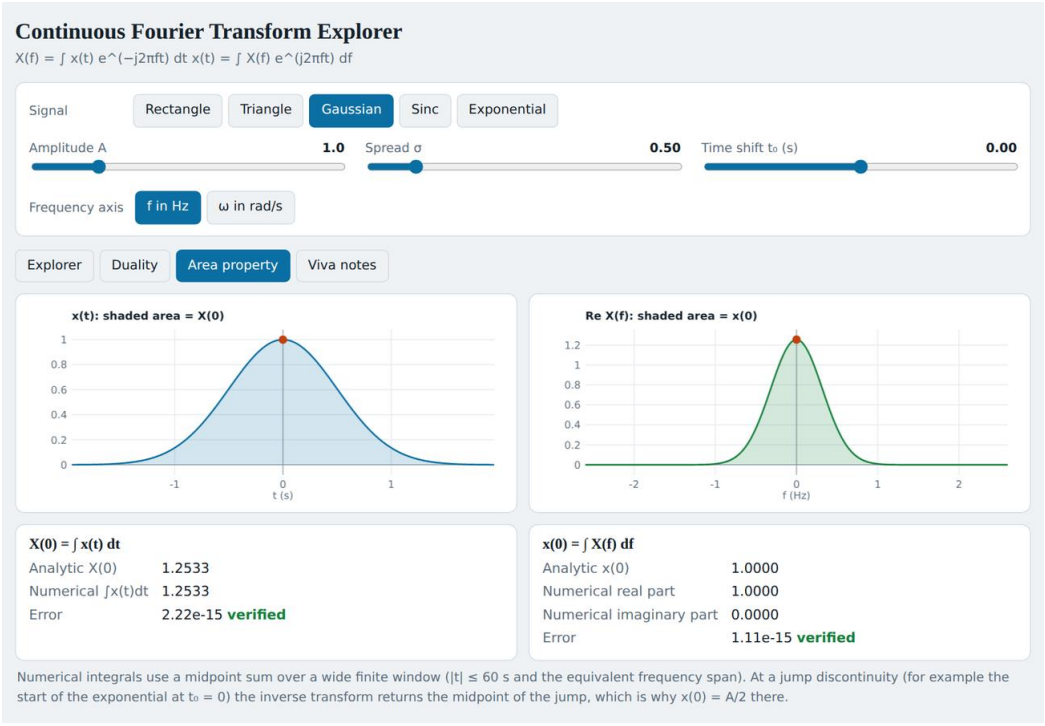


Figure 15. Area property, Gaussian (σ = 0.5). The area under x(t) equals X(0) = 1.2533 and the area under X(f) equals x(0) = 1.0000; both errors are about 10^{-15}.

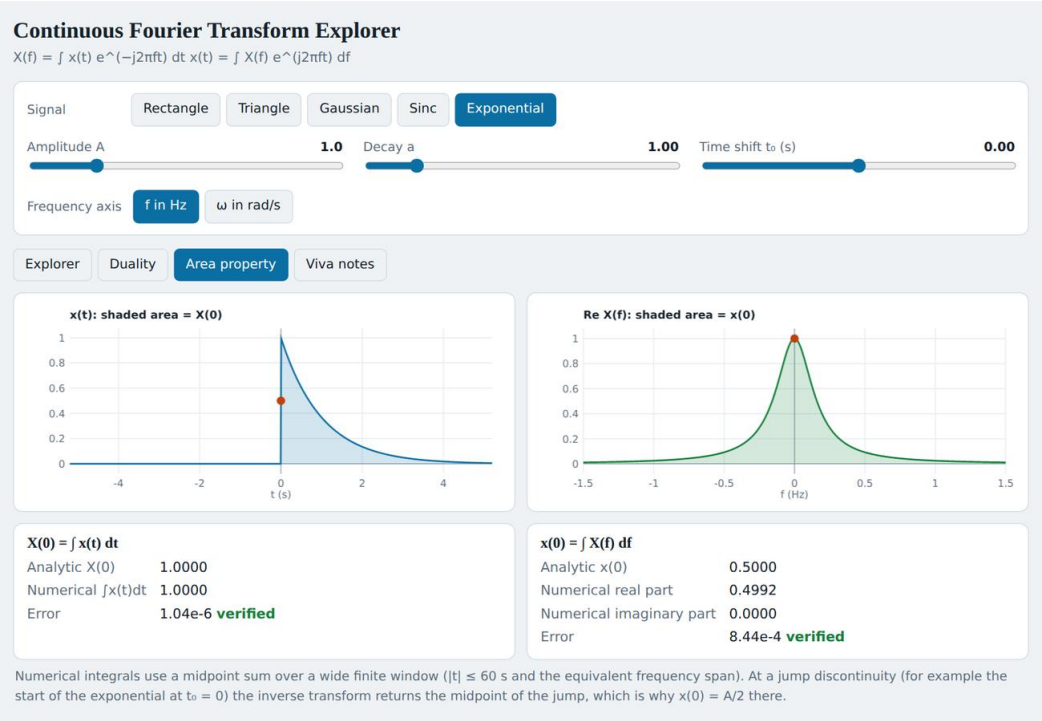


Figure 16. Area property, exponential ($a = 1$, $t_0 = 0$). The area under $x(t)$ equals $X(0) = 1$; the area under $X(f)$ is 0.4992, matching $x(0) = 0.5$, the midpoint of the jump at $t = 0$.

C.6 Values shown by the program

Table 8. Numbers read from the running program ($A = 1$, $t_0 = 0$).

Test	Closed form	Numerical	Error	Verdict
Duality, rectangle, Hz	x(−f) = rect(f)	FT of X(t)	RMS 1.76×10^{-2} (1.76 % of peak)	duality verified
Duality, exponential, rad/s	2π·x(−ω)	FT of X(t)	RMS 1.29×10^{-2} (0.21 % of peak)	duality verified
Area, Gaussian: X(0)	1.2533	1.2533	2.2×10^{-15}	verified
Area, Gaussian: x(0)	1.0000	1.0000	1.1×10^{-15}	verified
Area, exponential: X(0)	1.0000	1.0000	1.0×10^{-6}	verified
Area, exponential: x(0)	0.5000	0.4992	8.4×10^{-4}	verified